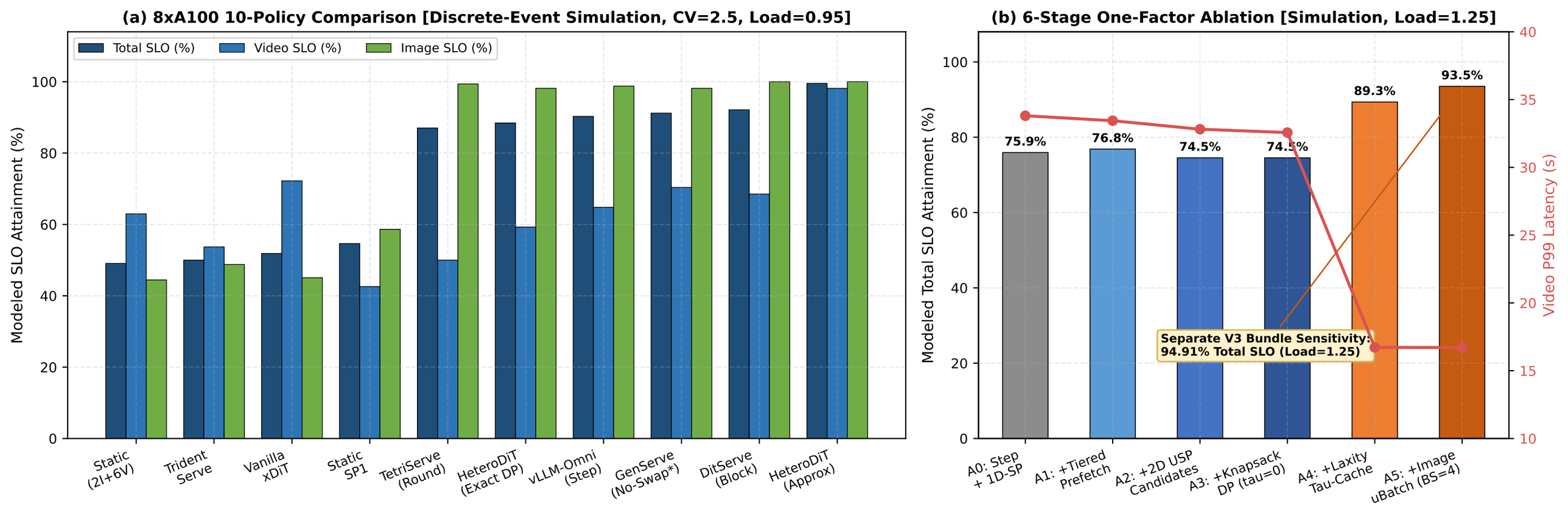
(a) 8xA100 10-Policy Comparison [Discrete-Event Simulation, CV=2.5, Load=0.95]
Total SLO (%)
Video SLO (%)
Image SLO (%)
Modeled SLO Attainment (%)
0
20
40
60
80
100
Static (2I+6V)
Trident Serve
Vanilla xDiT
Static SP1
TetriServe (Round)
HeteroDiT (Exact DP)
vLLM-Omni (Step)
GenServe (No-Swap*)
DitServe (Block)
HeteroDiT (Approx)
(b) 6-Stage One-Factor Ablation [Simulation, Load=1.25]
Modeled Total SLO Attainment (%)
Video P99 Latency (s)
0
20
40
60
80
100
10
15
20
25
30
35
40
75.9%
76.8%
74.5%
74.5%
89.3%
93.5%
Separate V3 Bundle Sensitivity:
94.91% Total SLO (Load=1.25)
A0: Step + 1D-SP
A1: +Tiered Prefetch
A2: +2D USP Candidates
A3: +Knapsack DP (tau=0)
A4: +Laxity Tau-Cache
A5: +Image uBatch (BS=4)